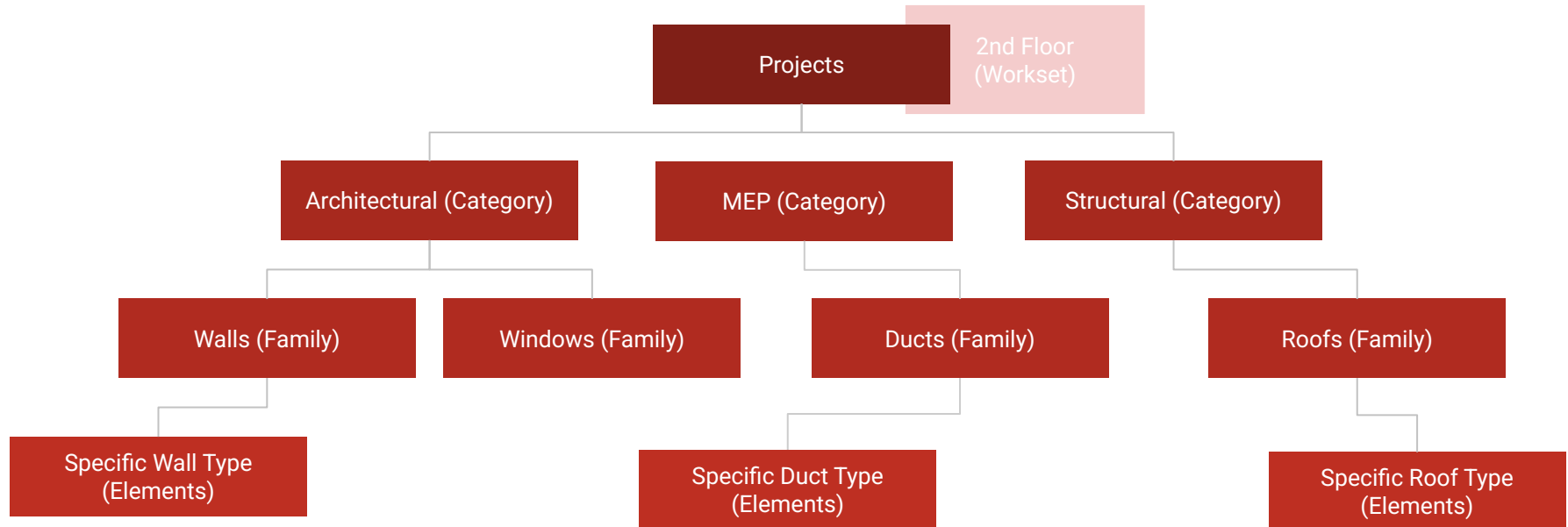


REVIT File Structuring



GENERAL STRUCTURES

- Object based approach → easy to manage components → decreased accuracy
- Best at building models which includes multiple systems
- Elements, families and categories vs parts, assemblies and subassemblies
- Data Organization:
 - All data from a project will be stored in the same file rather than individual components having their own file



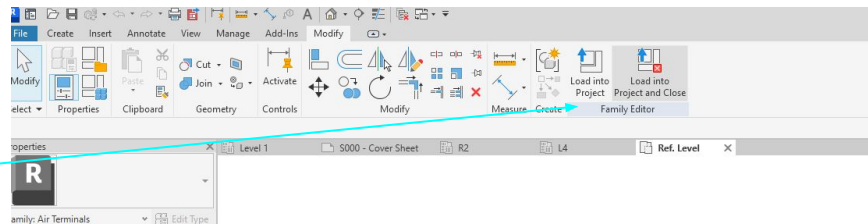
PROJECTS

- Projects are the entire building, site, or area which you are working on
- All data and elements are stored in the project file also known as the central file
- Projects consist of the project file, 2D or 3D views of the project data, sheets, families, and elements
- Sheets are collection of views organized for exporting
- Families are collections of related elements
- Elements are each singular component of the design



FAMILIES

- Use *Family Editor* to modifying existing elements or create new ones
- Families are an organized collection of similar elements, typically within the same category
 - I.E walls, windows, floors, plumbing
- There exist predefined families in Revit
- To add an element into a family, do one of the following:
 - Click the element to create and select in the desired family type in *Type Selector*
 - Select the family type in the *Project Browser* and drag to the drawing area
 - Select family type in the *Project Browser*, right click and select *Create Instance*
- To change the family type of existing elements:
 - Open a view that displays desired elements
 - Select 1+ elements in the same category and select the desired family type
 - OR Select the desired family type in *Project Browser*, right click, select *Match* and select each element in the drawing area
 - OR
 - Click Modify tab > Clipboard panel >  (Match Type Properties). In the view, select the element whose type you want to match. Then select elements to apply that type to.



To edit a family from a project

Do one of the following:

- Select an instance of the family in the drawing area, and click Modify | <Element> tab > Mode panel >  (Edit Family).
- Double-click an instance of the family in the drawing area.

Note: The double-click editing behavior is determined by the setting for the Family element type in the Double-click Options. See [User Interface Options](#).

MORE ON FAMILIES

- Finding instances of families in a project:
 - Open *Project View*
 - Within *Project Browser*, right click desired family and select EITHER:
 - Select All Instances > In Entire Project
 - Select All Instances > Visible in View
- Importing Family Types:
 - Family Editor > Insert tab > Import panel > Important Family Types
 - Select the family type file to import and click *Open*
- Deleting Family Types:
 - Families hosting other families, families in use, and system families may not be deleted
 - In *Project Browser*, select the family
 - Right click to delete or simply press delete

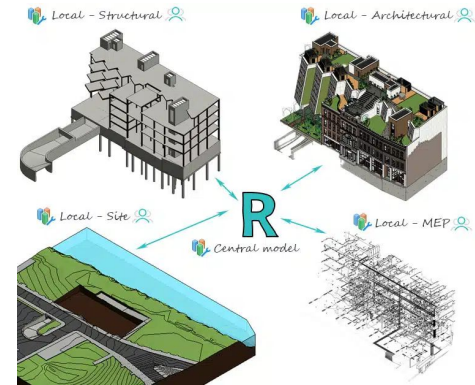
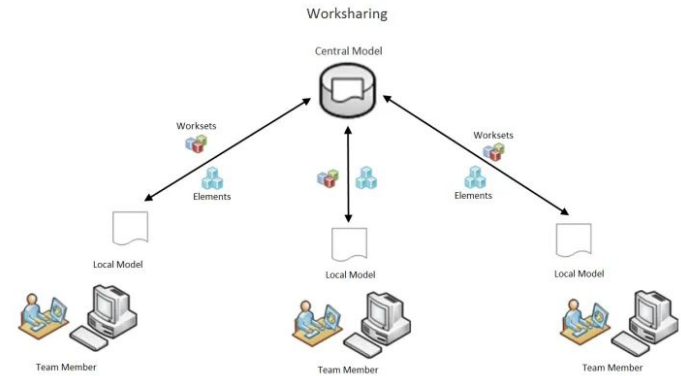
If you want to...	then...
change type properties	right-click a type, and click Type Properties. As an alternative, double-click a type, or select it and press Enter.
rename a family or type	right-click the family or type, and click Rename. As an alternative, select the type and press F2.
add a type from an existing type	right-click a type, and click Duplicate. Enter a name for the type. The new type displays in the type list. Double-click the new type to open the Type Properties dialog, and change properties for the new type.
add a new type	right-click a family, and click New Type. Enter a name for the type. The new type displays in the list. Double-click the new type to open the Type Properties dialog, and define properties for the new type.
copy and paste a type into another project	right-click a type, and click Copy to Clipboard. Open the other project and, in the drawing area, press Ctrl+V to paste it.
reload a family	right-click a family, and click Reload. Navigate to the location of the updated family, select it, and click Open.
edit a family in the Family Editor	right-click a family, and click Edit. The family opens in the Family Editor.
delete a family or type	right-click a family or type, and click Delete from Project. In addition to deleting the family or type from the project, this function deletes instances of matching types that exist in the model. Note: You cannot delete the last type in a system family.

Term/Concept	Definition
System Families	Part of the Revit environment used to create basic building elements, such as walls, roofs, and floors.
Loadable Families	Families that are: • created independently of the model, and loaded into the model as needed. • used to create installed building components like doors and fixtures, as well as annotation elements. • often hosted by system families. For example, doors and windows are hosted by a wall.
In-place Families	Custom elements that you create in the context of a model. Create an in-place family when your model needs unique geometry that you do not expect to reuse or geometry that must maintain relationships to other model geometry. Because in-place elements are intended for limited use in a model, each in-place family contains only a single type.
Modeling Families	Loadable families that represent real objects, such as doors, floors, or furniture. These families display in all views.
Annotation Families	Loadable families that are for annotation purposes, such as text, dimensions, or tags. These families do not have a 3D purpose and display only in the view in which you place them.
Category	Classification for families, for example doors, curtain walls, furniture, lighting fixtures, and so on. Families are grouped and sorted by category in the content libraries and in the Project Browser.
Family Types	Variations of the family element. For example, a family could be a door with a vision lite, and the types are 3 different sizes of that style door.
Instance Properties	Contain information related to a specific placed instance of the family element in a model. For example, instance properties for a door could include the sill height and the frame material. Changes made to instance properties affect only that instance of the family.
Type Properties	Contain information that applies to all instances of the same family type in the model. For example, type properties for a door could include the thickness and the width. Changes made to type properties affect all instances of the family created from that type.

WORKSETS

A collection of elements which are related through location or any other desired categorization (I.E. Interior, Exterior, Plumbing, etc)

- When you create a workset, you take ownership and therefore must make the workset editable for others to collaborate
 - Only one editor at a time
- Becoming the sole owner of a workset means borrowers must request to change any element
- Borrowing from a workset → taking or modifying an individual element from a workset
- It is recommended to work in your local copy of the central model and not make worksets editable → editing an element makes you the borrower of the element
- Synchronize with the central frequently to allow other team members to edit borrowed elements
- Use worksharing display modes to visually distinguish checkout status and model updates
- Performance (Why Use?)
 - Managing visibility
 - Reduce computational needs by closing worksets not worked on
- **Removing a workset:**
 - Must be editable to be deleted
 - When deleting, elements in workset can either be completely deleted or moved to a separate workset



HOW TO - WORKSETS

1. Creating Worksets

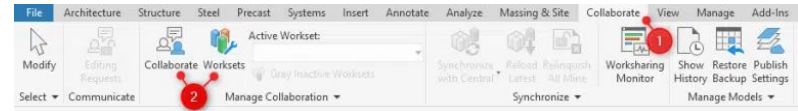
- Open *Collaborate Tab > Manage Collaboration > Worksets or Collaborate*
- Under Worksets dialog, click *New*
- Enter workset name

2. Changing visibility

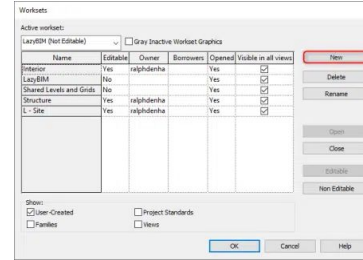
- Visible in all views* in *Visibility/Graphics* dialog for each workset to view complete project
- Deselect for irrelevant worksets

3. New worksets have option to be editable or non editable

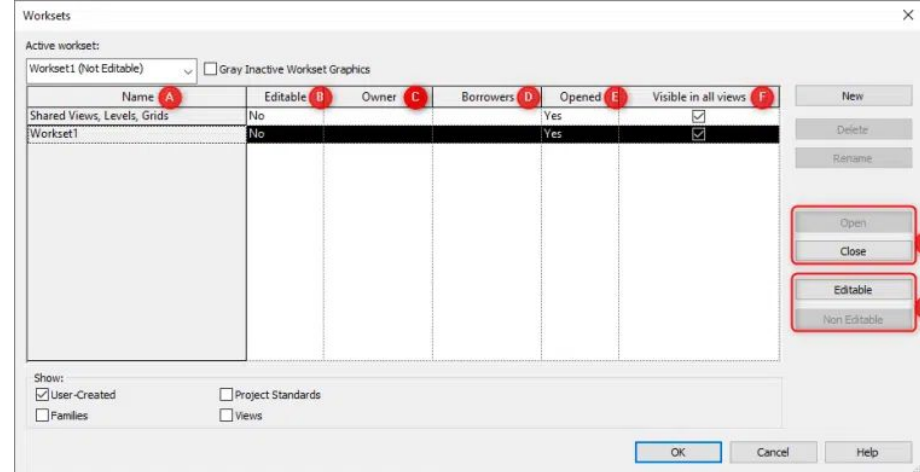
- If you wish to assign separate owners to each workset, each member must open a local copy of the central model and select the workset under *Worksets dialog* and select *Yes* under *Editable*



1a



1c



2a

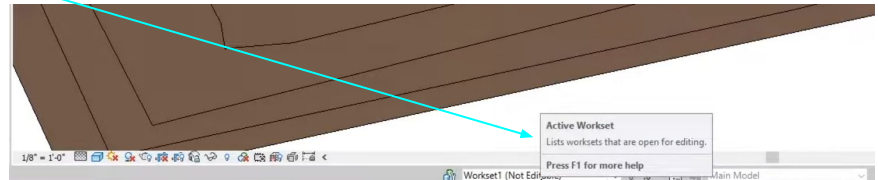
3

MORE ON WORKSETS

Terminology	Description
Worksharing	This design method allows multiple project members to work on the same model simultaneously.
Workshared file	A Revit file with Worksets enabled.
Collaboration	Multiple project members are working on the same project. The project members may be from other disciplines, such as architecture, construction, or mechanical. You can achieve collaborative work methods through worksharing and linked models.
Workset	A bundle of elements in a project with a specific ownership. When using Worksets in Revit, each discipline can be divided into separate Worksets. Tasks can be assigned accordingly and each individual can be held responsible.
Central file	This is the main file for workshared projects. The central file stores the present ownership details for all elements in the Revit project. Moreover, it serves as the central hub for all changes that are published to this file. Each Revit user works on a local copy and syncs with the central file.
Local file	This file is a copy of the Central file, where each Revit user works on their Workset. Project members should synchronize their changes periodically to the central model. This allows other members to see the changes as well.
Borrowing	This is a feature that allows you to edit an element that you do not have ownership of. When no users own an element, you get permission to borrow automatically. When someone else is editing an element in the project, they become the owner. You can request to borrow that element.
File-based Worksharing	This method allows you to create a workshared model stored at a network location.
Cloud Worksharing	This method allows you to store a workshared model in the cloud, such as the Autodesk Construction Cloud , for example.

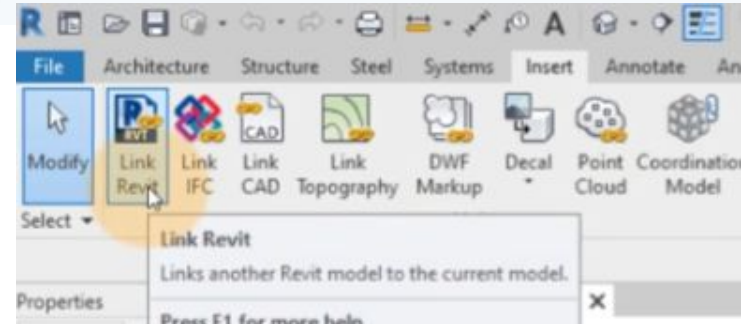
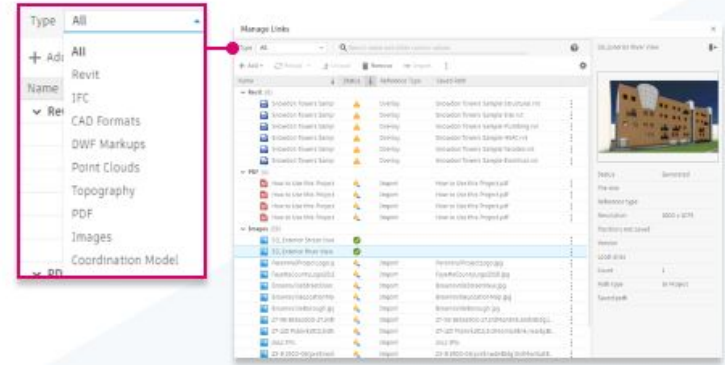
- Closing a workset hides it globally
- Worksets not opened could still be referenced but will not be loaded in
 - Saves memory and optimizes performance
- Visibility can be changed to hide or show only within the 3D or 2D models
- Visibility can be changed to hide worksets which are irrelevant to current task
- Changing the active workset changes where modifications and new elements are located
- Use the *Reload Latest* to update your copy of the project without changing the central model

Bottom of page



LINKED FILES

- Links are managed through “[Manage Links Dialog](#)”
- Link types: Revit, CAD, DWG, IFC, PDF, Point Cloud, Loads Analysis
- **Revit links**
 - Reference type should typically be “Overlay” to avoid links becoming connected
 - To recognize walls from links, Select “Room Bounding” within “Edit type”
- **Linked CAD files**
 - will update to reflect changes after reloading
 - Import units
 - Imported categories will not be added
 - To hide/change display of imported or linked files:
 - Modify | <File Name> tab » Import Instance panel »  (Query)
- **Orphaned Linked Elements:**
 - If the tag of an element in a linked mode is deleted or moved, they become orphaned
 - Orphaned tags will automatically rehost the element if previous host become available
 - Review through: Collaborate tab » Coordinate panel »  (Reconcile Hosting).



FILE COMPATIBILITY

- Revit supports imports and exports of these file types:
 - Revit files:
 - RVT, RFA, RTE, RFT
 - CAD:
 - DWG, DXF, DGN, ACIS (SAT), OBJ, 3DM, STEP, SKP, STL, AXM
 - Images:
 - BMP, PNG, JPG, JPEG, TIF
 - Other:
 - ODBC, HTML, TXT, gbXML

Revit does not support direct Creo files (both importing and linking)

Alternatives:

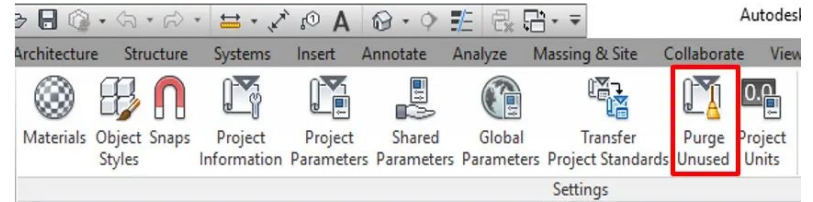
- Export Creo model as STEP, OBJ or STL
- Converting into compatible format before importing or linking
- Relationships, volume and geometries may not be accurate
- BIMDeX plugins: data exchange tool

GEOMETRIES

- When importing a file, the quality of the geometry may vary by file type, export settings, import settings and complexity of geometry
- Limitations:
 - Geometry in Revit models must stay within 10 miles/16 km of the internal origin to preserve accuracy
 - Move internal origin through moving a survey point BEFORE beginning a project to preserve accuracy
 - Overly complex geometry will impact Revit's performance
- Revit's geometry tool requires volumetric geometry

REDUCE BULK

- Purge Unused Objects:
 - All worksets must be opened to purge
 - Recommended to create backup file
 - Manage Tab > Settings > Purge Unused
 - Select and clear boxes to indicate which objects to purge, then select OK
- Using worksets can improve system performance
 - Ability to reference models without opening entire projects
 - Easily workshare without needing to load massive files
- Close unneeded worksets to reduce load on RAM



AUTODESK CONSTRUCTION CLOUD

- Limitations

- Max # of members granted permission to a folder: 400
- Max files per folder: 20,000
- Max file size: 5 TB
- Max pages in a PDF file: 2,000
- Max # of versions per file: 10,000
- Files which can't be downloaded: data exchanges and snapshots of Model Coordinations views saved to Docs

- Assemble

- Construction professionals can condition, query and connected BIM data to workflows such as design reviews, change management and value engineering
- Offline mode in Assemble iOS mobile app for no wifi on worksites
- Can use Color by Property to preserve color-codes

- BIM 360

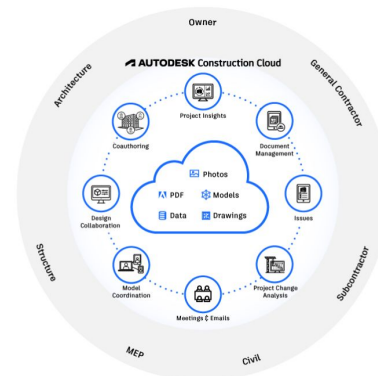
- Zoom integration

- BuildingConnected

- Use "J" key to jump between bid packages or companies
- Bidder List Templates can group preferred contacts and import them in bulk

- PlanGrid

- Create custom lists and self-assigned tasks



- Autodesk Docs connected drives in Revit can link and open files from the Autodesk Docs data source

- Install Desktop Connector and use the displayed Autodesk Docs connected drive
- Only RVT, RTE, and RFA files can be opened but click the above link to see what file types can be linked
- Recommended to be used to compliment Revit Cloud Worksharing workflows
- Considerations and Limitations
 - The connected drive does not immediately download the latest changes by other users
 - If two users open the same file, the last user to save will override the file
 - You will not be able to navigate into the connected drive contents if running Revit as an administrator
 - The connected drive cannot host workshared central models
 - You cannot load multiple family files into the connected drive

REVIT CLOUD WORKSHARING

Feature within Revit that allows multiple users to collaborate

Uploading a workshared model to Revit Cloud:

1. Open or create a model
2. Collaborate Tab > Manage Collaboration > Collaborate > In the Cloud
3. Select desired project folder
4. Select Initiate and close to continue
5. If notifications are allowed, you will receive an email when your model is successfully published to the cloud
6. If the model has linked files, migrate the links to Autodesk Docs

OR

1. Open model > Collaborate tab > Manage Collaboration > Worksets > OK
2. Click *Save As*
3. Select Cloud Model and save
4. Migrate links as needed

OR

1. Open model > Save As > Cloud Model
2. In the *Save as Cloud Model* dialog, find the desired project and folder in the cloud
3. Select *Save*
4. Select *Close* to continue
5. Migrate links as needed
6. Collaborate > manage Collaboration > Worksets
7. In the *Enable Worksharing* dialog, select each of the check boxes to confirm your understanding of the workflow
8. Click *Enable* and *OK*

AUTODESK CONSTRUCTION CLOUD

Cloud-based platform which integrates into multiple Autodesk tools

Uploading files and managing file versions:

1. Open project in Autodesk Docs
2. Select desired project folder
3. Click *Upload Files*

OR

1. Open Windows Explorer and select the desired file(s)
2. Drag files into the file list inside the project folder
3. Click *Done* to minimize while files are uploading
 - a. You are able to upload additional files during this time

- The ACC has the ability to open more than 100 files within a project at a time
 - [Supported file types](#)
- Features:
 - Measure, markup and work tools
 - Autosave progress
 - After changes, the version of the file/document will be updated and version history can be viewed

SOURCES

[Families - Essential Skill](#)

[Planning Revit Families](#)

[Revit Families](#)

[About the Different Kinds of Families](#)

[Find Instances of a Family Used in the Project](#)

[Complete Guide to Worksets](#)

[Creating Worksets](#)

[Make Worksets Editable](#)

[Manage Links Dialog](#)

[Linking files](#)

[Using Info from Other Apps](#)

[Layers in Linked and Imported Files](#)

[Linking CAD Files](#)

[About Importing or Linking CAD Files](#)

[About Imported Geometry in Mass or Generic Model Families](#)

[Purge Unused Objects](#)

[Performance \(Why Use?\)](#)

[Limitations](#)

[Supported file types](#)

[Upload Files and Manage File Versions](#)

[Autodesk Construction Cloud Product Tricks](#)

[Performance \(Why Use?\)](#)

[Revit Cloud Worksharing](#)

[Best Practices: Working on a Team Project](#)

[Working with Cloud Models](#)

[About Autodesk Docs Connected Drive Integration with Revit](#)